Lecture 22 Practice Questions and Solutions

Question 1 (q1_sol.py)

SFU term codes are 4-digit numbers, where each number has a specific meaning. For example, the term code 1267 means Fall 2026 because:

- the first digit is a 1 (for this century)
- the next two digits are 26, the last two digits of the year 2026
- the last digit is a 7, which is the semester code for Fall

Using your semesters dictionary, write code that takes a semester name and a year and prints the term code. It should read in the semester name, the year, and then print the term code. You do *not* need to repeatedly ask the user, i.e. it can stop after the printing the term code.

3 Sample Runs

```
Enter semester: Fall
Enter year: 2026
Fall 2026 term code: 1267

Enter semester: Spring
Enter year: 2027
Spring 2027 term code: 1271

Enter semester: Summer
Enter year: 2026
Summer 2026 term code: 1264
```

Use this dictionary in your program:

```
codes = {'spring': 1, 'summer': 4, 'fall': 7}
```

Sample Solution

```
codes = {'spring': 1, 'summer': 4, 'fall': 7}

# get the input
semester = input('Enter semester: ').strip().lower()
year = input('Enter year: ').strip()

# process the input: construct the term code
term_code = '1' + year[-2:] + str(codes[semester])

# print the result
print(f'{semester.capitalize()} {year} term code: {term_code}')
```

Question 2 (q2_sol.py)

Write a program that repeatedly asks the user to enter a term *code number* and the prints the corresponding semester name. It should stop when the user types <return>, and print a helpful message if an invalid code is typed. Use a dictionary to efficiently store the semester codes.

Here are the semester codes:

```
Semester   Code
---------------
 Spring      1
 Summer      4
 Fall        7
```

Sample Run

```
Enter semester code: 4
4 is Summer

Enter semester code: 8
8 is invalid

Enter semester code: 1
1 is Spring

Enter semester code: <return>
All done!
```

Sample Solution

```
codes = {1: 'Spring', 4: 'Summer', 7: 'Fall'}

while True:
    code = input('Enter semester code: ')
    if code == '':
      break
    code = int(code.strip())
    if code in codes:
        semester = codes[code]
        print(f'{code} is {semester}')
    else:
        print(f'{code} is invalid')
    print()

print('All done!')
```